

PPA Exposure Model

Methodology

1. What this model does

This tool estimates the **counterparty credit exposure** arising from a fixed-for-float Power Purchase Agreement (PPA) over its full lifetime of five years. The model simulates thousands of possible future power price scenarios using Monte Carlo simulation, revalues the PPA under each scenario at monthly intervals, and extracts the exposure profile. The output is used for credit limit setting, portfolio monitoring, and regulatory capital computation.

2. Price dynamics

Power prices tend to **revert to an equilibrium level** driven by supply and demand fundamentals, and they exhibit **seasonal patterns**. The model captures both features using the Schwartz one-factor framework, a standard approach in energy risk management. The power price follows a mean-reverting process in log-space (Ornstein-Uhlenbeck):

$$dX_t = \kappa(\mu(t) - X_t) dt + \sigma dW_t$$

where $X_t = \ln(S_t)$, κ controls how fast prices revert to equilibrium, $\mu(t)$ is the seasonal mean level, and σ is the annualized volatility. The seasonal component follows a sinusoidal pattern calibrated to the European power market.

Model parameters

Parameter	Symbol	Description	Default
Mean-reversion speed	κ	Speed of reversion to equilibrium	1.5
Long-run mean	μ	Equilibrium price level (EUR/MWh)	52.0
Volatility	σ	Annualized log-price volatility	35%
Seasonality amplitude	A	Seasonal swing in log-price	0.15
Seasonality phase	φ	Phase shift (0 = winter peak)	0.0

3. Forward curve

The model supports two forward curve modes, selectable in the application:

Model-implied. The forward curve is derived analytically from the parameters. The model-implied forward price for delivery at time T , given spot S_t , is computed from the conditional moments of the OU process. This mode is useful for scenario analysis and stress testing where you want to explore different parameter regimes.

Market-anchored. The user provides an observed forward curve. The simulation generates stochastic deviations around it while ensuring that the expected simulated price at each tenor matches the market forward. This is the standard production approach — it separates the market's view of fair value (the curve) from the model's role (generating uncertainty around it).

4. PPA valuation

The PPA is modeled from the **generator's perspective** (fixed price receiver). At each time step in the simulation, the Mark-to-Market is the present value of all remaining cash flows:

$$MtM(t) = \sum (K - F(t, t_i)) \cdot Q_i \cdot DF(t, t_i)$$

where:

- **K** = fixed price (EUR/MWh)
- **F(t, t_i)** = model-implied forward at time t for delivery at t_i (no look-ahead bias)
- **Q_i** = delivered volume = Capacity × Capacity Factor × Hours per period
- **DF(t, t_i)** = continuous discount factor

5. Exposure metrics

The following metrics are computed from the simulated MtM distribution at each time step across all Monte Carlo paths:

Metric	What it tells you	Used for
EE (Expected Exposure)	Average credit exposure at each point in time	Portfolio monitoring, management reporting
PFE (Potential Future Exposure)	Worst-case exposure at a given confidence level (e.g., 99%)	Credit limit setting
EPE (Expected Positive Exposure)	Time-weighted average of EE over the contract life	CVA calculation
Effective EPE	Non-decreasing version of EPE (regulatory definition)	Regulatory capital (Basel III / SA-CCR)
Peak PFE	Maximum PFE across all time horizons	Credit limit allocation

The **Peak PFE** is the key output for credit risk management, it represents the maximum counterparty credit limit required to support this PPA.

6. Model validation

Convergence analysis. Monte Carlo estimates are subject to sampling error. The model runs simulations with increasing path counts (500 to 50,000) and tracks the stability of Peak PFE and the standard error of the mean exposure. A converged model shows PFE stabilizing within a narrow band as paths increase.

Distribution diagnostics. The MtM distribution at each time step is analyzed for skewness, excess kurtosis, and normality (Jarque-Bera test). A QQ plot compares the empirical distribution against a normal reference. Commodity-linked exposures typically show fat tails, which means PFE at high confidence levels is sensitive to distributional assumptions.